Gap-filling Procedures
Supporting Document `IndigoCarbon_US-1_2023_0031`

June 28, 2024

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1 Gap-filling with Extrapolation and Default Replacement

1.1 Introduction

Extrapolation and default replacement are the processes used to gap-fill and resolve QA/QC errors related to missing data or out-of-range attributes. (See `IndigoCarbon_US-1_2023_0032` for details regarding QA/QC.) The use of extrapolation to gap-fill both qualitative and quantitative data values was approved by the Registry in April 2021 (see `IndigoCarbon_US-1_2023_0067a`) on the basis that grower-provided data for 'similar fields' per SEP v1.1, 6.1 data hierarchy, level 1c (here defined as the same field in other years) will be more accurate for that field than generalized default values taken from alternative data sources such as the National Agricultural Statistics Service (NASS; see `IndigoCarbon_US-1_2023_0069` for details), published literature, or similar sources. Gap-filling with default replacement is attempted only when extrapolation has failed to resolve QA/QC errors. It is conducted in accordance with the SEP v1.1, 6.1 data hierarchy, level 2, which allows gap-filling to be done using "... values derived from agricultural census data or other sources...".

Figure 1 illustrates how an error from QA/QC propagates through these processes until it is either resolved and the field is sent through to model input creation, or not resolved, and the field is prevented from being sent to the model or quantified for crediting. Restrictions are made to gap-filling procedures to ensure that program eligibility remains unaffected by gap-filling. This is also checked explicitly with the re-run of QA/QC after gap-filling is finished.

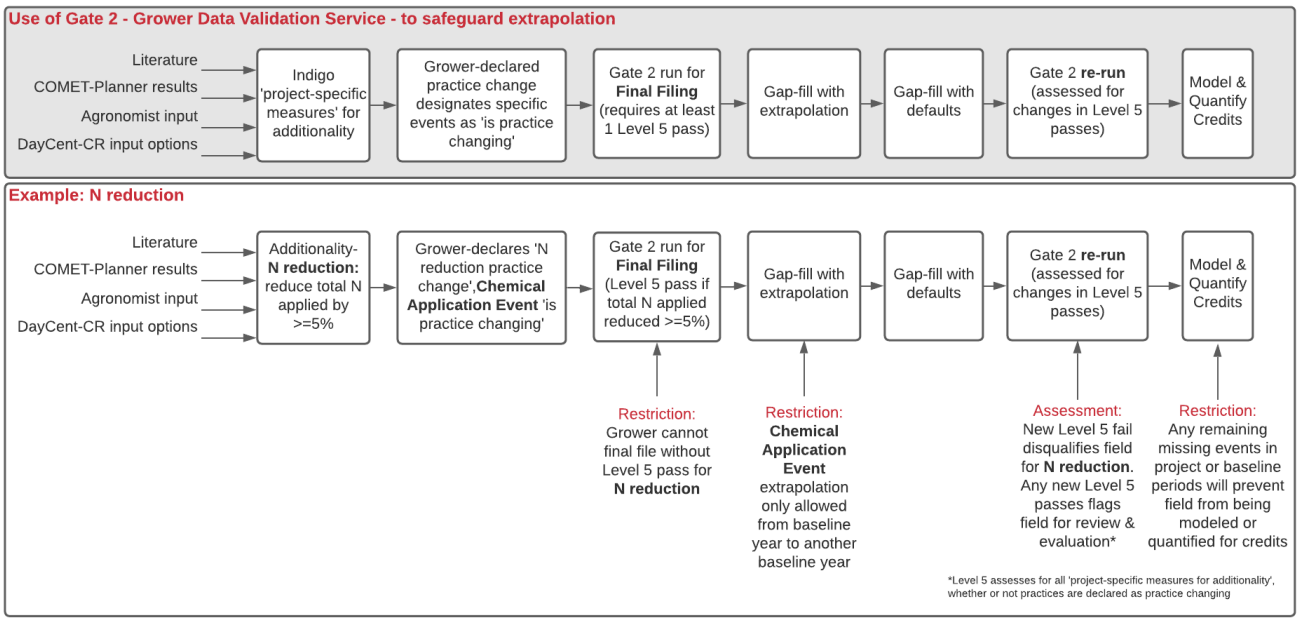


Figure 1: Data flow through Gates 2 and 5, representing the Validation Service and Gap-filling, respectively.

Errors detected during QA/QC are first considered for extrapolation, where grower data for that field in other years that meets specific requirements may be used to fill gaps and replace out-of-range values (see *Extrapolation* section). Because of the specific requirements on data used to extrapolate, including the use of practice change designation to restrict criteria allowing extrapolation to be completed (see Figure 1), example box; see *Extrapolation* section), extrapolation may not always be successful. In this case, the error is passed on for attempted default replacement, again with restrictions based on whether it is associated with a practice changing event (see *Default Replacement* section). Default replacement may succeed or fail, depending on the availability of external data sources used in

default replacement and/or the circumstances of the error. Successful resolution of errors will enable a field to continue through the crediting process, to modeling and credit quantification steps. Any remaining errors unresolved by both extrapolation and default replacement will fail and thus prevent a field from being modeled or quantified for credits.

Gap-filling is attempted only after program eligibility is assessed for qualifying practice changes. Specifically, the field must pass the SEP Performance Standard Test for Additionality (PST; see SEP v1.1, 3.4.1) without any form of gap-filling. Program eligibility is then reassessed after gap-filling by re-running Gate 2 Level 5. Figure 1 illustrates the order of operations for these steps using an example of a field with an increased cover crop duration practice change. In this case, cover crop planting and/or termination dates are allowed to be extrapolated within the baseline using only other baseline events (See *Principles of Extrapolation*). Similarly, default replacement is only allowed on non-practice changing events and attributes. When Gate 2 Level 5 is re-run after gap-filling is complete, field eligibility is reassessed. Any field that has lost eligibility will be excluded from crediting. While eligibility must remain unchanged with gap-filling for a field to proceed with crediting, it is expected that gap-filling procedures may impact crediting outcomes on individual fields in both positive and negative directions, for example due to the use of specific agronomic parameters or timing of events. A final assessment is performed evaluating the net impact on credits caused by gap-filling, see `IndigoCarbon_US-1_2023_0075` for full analysis).

1.2 Principles of Extrapolation

Extrapolation is attempted before default replacement on the basis that grower data for the same field in a different year growing the same crop will be more accurate than regional data sources like NASS or generalized data sources like peer-reviewed literature. To help ensure extrapolation does not alter eligibility, restrictions are made to the extrapolation of practice-changing events and attributes. For example, extrapolation is allowed on events and attributes associated with practice changes only in the baseline historic period and only using other events that qualify for extrapolation in the baseline historic period. Further, extrapolation of baseline events and attributes to with-project scenarios is allowed, and vice versa, but only for missing information in non-practice changing designated events. Sharing extrapolated data between baseline and with-project scenarios will neutralize practice changes at a project level, and thus have a conservative effect on credits.

1.3 Principles of Default Replacement

Default replacement draws from external data sources to remedy missing values or replace values detected as out-of-range by QA/QC. Default replacement is conducted in accordance with the SEP hierarchy for both qualitative and quantitative data (SEP v1.1, 6.1 data hierarchy, level 2). External data sources used in default replacement are documented in `IndigoCarbon_US-1_2023_0069` and cited as necessary to this document.

When errors arise due to conflicts between grower-provided values and QA/QC data sources such as NASS and the Cropland Data Layer (CDL; see `IndigoCarbon_US-1_2023_0069`), grower-provided values are retained, on the basis that the QA/QC data sources are generalized and less accurate. This principle, which aligns with CAR's written guidance (`IndigoCarbon_US-1_2023_0067a`), means that in some cases no default replacement will take place and an error will be considered automatically resolved (examples include *Cash Crop Type* and *Tillage Date* logic detailed in the sections below).

To ensure default replacement does not alter eligibility, default replacement is restricted to events and attributes that are not associated with practice changes. This promotes the use of data associated with a higher level of the SEP v1.1, section 6.1 data hierarchy (i.e., grower-provided values) in the modeling of practices that are most critical for quantifying emission reductions. To further ensure the effects on credits are unbiased or conservatively biased at the project scale, default replacement is performed identically on baseline and with-project events and attributes.

In cases where numeric data sources could not be found, literature, guidance from agronomic subject matter experts (SMEs; see `IndigoCarbon_US-1_2023_0032` for SME credentials), or results from model experiments were used to determine appropriate values to be used in default replacement. Default replacement approaches are also designed to be consistent with QA/QC data sources used to create QA/QC range checks (see Turnstile 3 description in `IndigoCarbon_US-1_2023_0032`), and software constraints.

2 Extrapolation

2.1 Extrapolation logic

The basic logic of extrapolation is:

1. Check for years with a matching cash crop
2. Use the hierarchy in 1.2 below to determine which matching cash crop year to use for extrapolation
3. Extrapolate the needed information (copy from one year to another)
4. Repeat as needed, progressing through the hierarchy in 2.1.1 for whole years, or 2.1.2 for events and attributes, until the field has all data filled or there are no more matching cash crop years available.
5. If events or attributes for the field remain unremedied, progress to default replacement

2.1.1 Criteria to extrapolate entire missing years

Many fields are missing entire years of data and thus contain no positive or negative events (figure 2). If an event is marked as "Did not occur" in the Carbon UI, registering it as a negative event within the software, it does not need to be gap-filled.

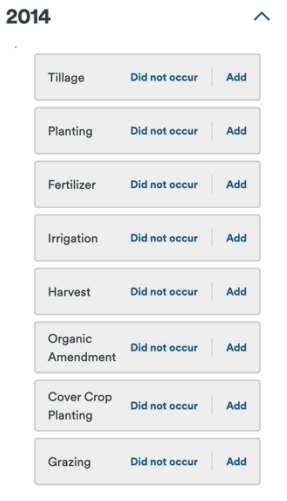


Figure 2: Example of missing data years.

Both positive and negative events can be extrapolated for a given year (e.g. fertilizer application is a positive event, while not tilling is an example of a negative event), and thus entire years may be extrapolated or copied over into missing years with a matching cash crop. However, entire years may be extrapolated only if both years occur in the baseline (see 2.1.2 below, since extrapolated years will necessarily contain practice-changing events). The general logic for selecting entire years with which to extrapolate is as follows:

a) Extrapolated years must have a matching cash crop

- CDL is used to determine the cash crop type for the missing year
  - If CDL is unavailable for that year, whole-year extrapolation is not allowed. Individual event or attribute extrapolation may be attempted (see 2.1.2)
- The cash crop type must be a direct match, and must be supported for crediting
- If multiple cash crops are planted that year (e.g., winter wheat followed by soybeans), a matching year must also contain both of those cash crops

b) Whole-year extrapolation is only allowed between baseline years

c) The closest year in time meeting the above criteria is selected

d) If there are no complete, error-free years with matching cash crop available then individual event and/or attribute extrapolation is attempted as described in 2.1.2.

2.1.2 Criteria to extrapolate event or attribute data

In the cases of missing or out-of-range events or attributes, eligible events and attributes for extrapolation are selected according to the following logic.

a) Extrapolated events and attributes must be associated with a matching cash crop

- As needed, CDL is used to determine the cash crop type for missing years
- The cash crop type must be a direct match, and must be supported for crediting
- If multiple cash crops are planted that year (e.g., winter wheat followed by soybeans), a matching year must also contain both of those cash crops.

b) Determine which matching event or attribute to extrapolate, prioritized as follows:

- For non-practice changing events and/or attributes
  - Match the baseline/project status (i.e. delineated by coming before or after the earliest practice change on that field)
    - Choose error-free, non-missing event or attribute from a year with no other errors or missing data that year
    - Choose error-free, non-missing event or attribute from year with other errors or missing data that year
  - Un-matched baseline/project status
    - Choose error-free, non-missing event or attribute from year with no other errors or missing data that year
    - Choose error-free, non-missing event or attribute from year with other errors or missing data that year
- For practice-changing events and/or attributes
  - Match the baseline status (i.e. these may only be extrapolated in the baseline from other baseline events)
    - Choose error-free, non-missing events or attributes with matching baseline status from a year with no other errors or missing data that year
    - Choose error-free, non-missing events or attributes with matching baseline status from a year with other errors or missing data that year

c) If multiple years end up meeting the requirements for selection per this logic (e.g., two baseline years of corn are available to fill in a missing corn year in the baseline), the year closest in time is selected.

d) Events that have been remedied with extrapolation may not also be extrapolated to other events.

e) Events may only be extrapolated if they occur on the entire field.

f) Instances of negative planting dates that are missing harvest dates are remedied by filling the missing harvest date with a negative harvest date.

g) When extrapolating a winter crop (e.g., winter wheat), years with matching crops are selected based on the planting date, not the harvest date. In these cases, the harvest date is added to the following year based on the original (extrapolated) harvest date for that crop. When the harvest date is added, it must not overlap with the next growing season (i.e. planting date). If it does, the harvest date is placed 1 week before any tillage or planting events in that next year.

2.1.3 Extrapolating Dates

Dates other than planting or harvest have additional logic applied. If an entire year is missing, fill in all dates as exact copies with the current year. When extrapolating a date that is not within an entire missing year, the following logic is applied:

- Determine if the extrapolated event date was closest to planting or harvest
- Calculate the offset before or after that planting or harvest event
- Create a new extrapolated event with the same offset in the year of the missing event
  - e.g., if tillage occurred 7 days before planting in the year being extrapolated from, the tillage event must be placed 7 days before planting in the year of the missing event

Example:

- A tillage date is extrapolated from 2021 to 2019 due to a missing tillage event in 2019. There are no other missing events in 2019.
- Tillage in 2021 occurred 7 days before a May 10th planting of corn. Therefore, the extrapolated tillage date in 2019 will be placed 7 days before planting in 2019.
- In 2019, corn was planted on April 20th, so the tillage date is filled in as April 13th, 2019.

2.1.4 Quality Assurance

Additionality is re-checked (Gate 2, Level 5) after extrapolation is performed to ensure no change has occurred to prior additionality determinations. Losses of additionality are accepted as conservative, while any gains to additionality are flagged and investigated to ensure there is no project-wide increase in credits due to extrapolation.

3 Default replacement

3.1 Conversion of missing events to positive or negative

The first step of default replacement is to determine if any new events must be created to remedy cases where a grower has not specified if an event has happened or not, aka "missing events". In these cases, the QA/QC workflow will register a missing event error which is then remedied with the missing event logic described below. For a given event type, the logic determines if the missing event should be converted into a positive event (i.e., assumed to have occurred) or a negative event (i.e., assumed to have not occurred). If a missing event is converted to a positive event, each attribute associated with that event is filled using the logic in the section *Gap-filling of Attributes*. Negative events are ignored in quantification as they indicate the absence of agronomic practice.

3.1.1 Missing Planting and Harvest events (i.e. "Planted Periods")

For missing event errors related to Planting, Cover Crop Planting, Harvest, and Termination events, the logic below (Figure 2) is used to determine if new positive events for planting and harvest should be created. The logic is based on a software constraint that requires planting and harvest (or termination) events be paired for a given cash crop type or cover crop species. (See Gate 2, Level 2 description in `IndigoCarbon_US-1_2023_0032`.) Pairs of planting and harvest events are referred to as "planted periods".

Logic

- Check the correct reference data source for the given event type.
  - Missing planting and harvest events for cash crops will use CDL (see `IndigoCarbon_US-1_2023_0069`).
  - Missing cover crop planting and termination events will use Indigo's Remote Sensing model for cover crop detection (see `IndigoCarbon_US-1_2023_0055`).
- Determine if there is a conflicting negative planting or harvest / termination event.
  - If yes, create a negative event (giving deference to the grower-provided response).
  - If no, determine if the reference data source indicates a planted period is expected.
    - If no, create a negative event.
    - If yes, determine if a matching planted period already exists
      - If yes, create a negative event.

- If no, determine if a new planted period will plausibly fit in the sequence of agronomic events
- If no, create a negative event
- If yes, create a new planted period.

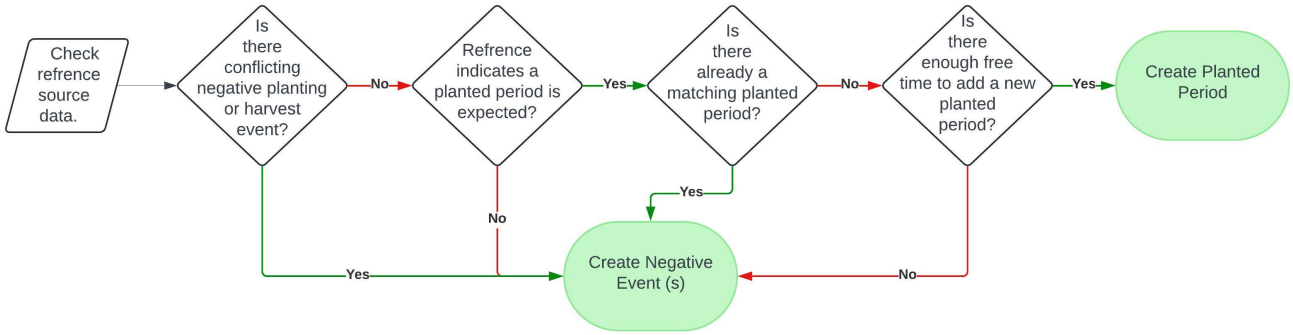


Figure 3: Logic to determine if new planting or harvest events should be created.

3.1.2 Missing Chemical Application Events

Logic

- Query the NASS Quick Stats database (see `IndigoCarbon_US-1_2023_0069` for details) for an N fertilizer application rate using cash crop type and state in which the field is located.
- If the state is missing from the database, use a proxy state (Table 1), defined based on the similarity of LRR composition within the state.
  - For example, GA is ~15% LRR N, 20% LRR T, and 65% LRR P. The best proxy state for GA is NC, which is ~15% LRR N, 25% LRR T, and 60% LRR P.
  - The proxy state should be chosen in rank order per Table 1: if A is available, use A; else if B is available use B; else if C is available use C.
- If a value is returned, create a positive Chemical Application Event.
- Else, create a negative event.

Rationale

- The NASS Quick Stats database provides average fertilizer rates for a wide range of major field crops of the US, including legumes such as soybeans. It is assumed that if there is no NASS-provided rate it is more accurate to assume that no fertilizer was applied.
- Agronomic SMEs indicate that fertilizing cover crops is rare, thus it is assumed that cover crops are not fertilized unless the grower indicates otherwise.

Table 1: Proxy states to use when the intended state is missing from the NASS dataset.

State	Proxy State A	Proxy State B	Proxy State C
AR	AL	TN	LA
CO	WY	MT	NE
GA	NC	SC	AL
IL	IA	IN	MO
IN	OH	IL	MO
IA	IL	IN	MN
KS	NE	OK	TX
KY	TN	WV	AR

Table 1: Proxy states to use when the intended state is missing from the NASS dataset.

State	Proxy State A	Proxy State B	Proxy State C
LA	MS	AR	AL
MN	WI	IA	IL
MS	LA	AR	AL
MO	IN	IL	OH
NE	KS	SD	OK
NC	GA	SC	VA
ND	SD	MT	MN
OH	IN	MO	IL
OK	KS	TX	NE
SC	GA	NC	VA
SD	NE	ND	MT
TN	KY	AL	NC
TX	OK	KS	LA

3.1.3 Missing Organic Amendment Application Events

Logic

- Check to see if organic amendments have ever been applied by this grower, on any field and at any point in time.
- Query the Census of Agriculture (Census; see `IndigoCarbon_US-1_2023_0069` for details) by county for the acres of cropland and pastureland on which animal manure was applied.
- If the grower has not otherwise applied organic amendments, and if the Census query indicates that $< 10\%$ of acres in the given county has received manure, create a negative event.
- Else, create a positive Organic Amendment event.

Rationale

- Use of the Census of Agriculture (see `IndigoCarbon_US-1_2023_0069`) data is consistent with Level 2 of the SEP v1.1, 6.1 data hierarchy for quantitative data.
- Counties where a high percentage of acres are treated with manure are assumed to be areas with a high concentration of confined animal feeding operations where growers will have incentives to apply manure to their operations, so in these counties a positive event is always created regardless of the grower's history.
- Agronomic SMEs report that organic amendment applications are primarily driven by local availability of manure, either from the grower's own livestock or from a nearby confined animal feeding operation. In counties where few acres are treated with manure, a grower who has not reported amendments elsewhere on their operation is likely not to have amended the field in question either.

3.1.4 Missing Tillage Events

Logic

- Find all the crop growing periods that overlap the window missing a tillage event.
- For each overlapping growing season:
  - Check whether other tillage events exist (i.e., outside the window flagged as missing).
  - If none are found, create a positive tillage event.
    - If a grower-reported negative tillage event conflicts with the default tillage date of the positive event, then create a negative event instead.
  - Otherwise, create a negative event.

Rationale

- Every crop growing season is assumed to have contained at least one tillage event unless the grower explicitly reports negative tillage events. The logic given here is simply to detect tillage schedules that are offset from the time windows checked for tillage by the quality control logic; once a whole crop growing season is confirmed to be missing tillage, a positive event is always created.

3.1.5 Missing Irrigation Events

Logic

- Query the MIrAD-US dataset (see `IndigoCarbon_US-1_2023_0069` for details) for irrigation status using the centroid of the field. Query the most recent year of MIrAD-US data occurring prior to the event.

Rationale

- The CAR Grassland protocol (https://www.climateactionreserve.org/how/protocols/grassland/) uses MIrAD-US to provide estimates of irrigated agricultural area in the U.S.
- Agronomic SMEs support the idea that if irrigation is established on a field at some point in the past, it will continue to be irrigated after that date.

3.2 Default Replacement of Attributes

Attributes undergoing default replacement are replaced with a default value according to logic outlined in this section. Because replacement of one attribute may depend on replacement of another attribute, irrespective of the attribute's event, a set of dependencies arises, and a specific order of replacement must be followed. The order of default replacement of attributes is illustrated in Figure 4, where each layer represents a chronological step in the sequence of default replacement. Dependent attributes will always occur in an earlier step in the sequence.

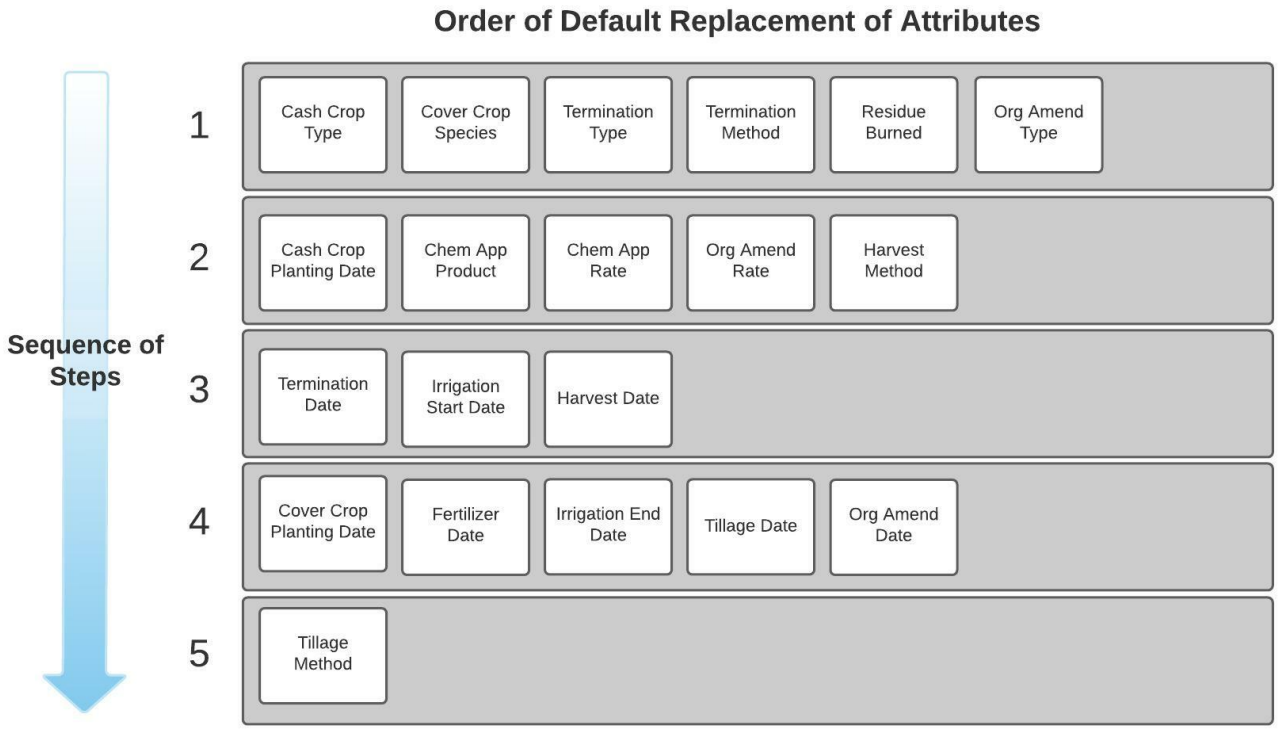


Figure 4: Order of default replacement of attributes.

The following section lists for each attribute the logic, rationale, specific dependencies with other attributes, and descriptions of any supporting model analyses used to derive the default replacement logic.

3.2.1 Cash Crop Type

Logic

- For new planted periods created from missing events, this attribute is replaced using the crop type returned by CDL (see `IndigoCarbon_US-1_2023_0069` for details).
- For existing planted periods this attribute is not replaced.

Rationale

- The use of CDL in replacing cash crop type for newly created planted periods is consistent with requirements for qualitative data in the SEP v1.1, 6.1 data hierarchy (Section 6.1, pp 70).
- For existing planted periods, errors associated with cash crop type are based on QA/QC checks against CDL and indicate a conflict with the grower-provided answer for the crop type that was planted. However, grower-provided values are prioritized over alternative gap-filling data sources including CDL, thus cash crop type is not replaced in these instances.

Attribute Dependencies

- None

3.2.2 Cash Crop Planting Date

Logic

- Check NASS Usual Planting Dates (see `IndigoCarbon_US-1_2023_0069` for details) using the associated cash crop type.
- For crop/state combinations that are missing, use proxy states (Table 2 presented in Missing Chemical Application Events section).
- If no result is obtained or too many results are obtained, fail.
- Convert the Usual Planting Date to day-of-year in the given year being gap-filled.

Rationale

- The use of NASS Usual Planting Dates conforms with Step 2 of the data hierarchy for quantitative data in the SEP, v1.1 (Section 6.1, pp 70).
- The approach is consistent with the QC ranges developed for planting and harvest events described in `IndigoCarbon_US-1_2023_0032`.

Attribute Dependencies

- Cash Crop Type

3.2.3 Cover Crop Species

Logic

- For new planted periods created from missing events, this attribute is replaced using "CEREAL RYE".
- For existing planted periods this attribute is not replaced.

Rationale

- The use of "CEREAL RYE" for replacement of cover crop species in newly created planted periods is supported by the SARE National Cover Crop Survey (Sustainable Agriculture Research and Education (SARE), 2020) which lists Cereal Rye as the most common cover crop species used in the US. The use of Cereal Rye is further supported by a 2021 USDA Economic Research Service report entitled *Cover Crop Trends, Programs, and Practices in the United States* (Wallander et al. 2021).
- For existing planted periods, errors associated with cover crop species are based on QA/QC checks against CDL and indicate a conflict with the grower-provided answer. However, grower-provided values are higher priority than alternative gap-filling data sources including CDL, thus cover crop species is not replaced. Further, because

CDL typically returns only one crop per year, it typically returns a cash crop and not a cover crop, suggesting that the errors are misguided.

Attribute Dependencies

- None

3.2.4 Cover Crop Planting Date

Logic

- This attribute should be replaced with the day after the harvest or termination date of the previous cash crop.

Rationale

- Planting day-after-harvest is agronomically reasonable and helps ensure sufficient time to germinate prior to freezing.
- Due to model limitations, it is not possible to represent cover crops planted into cash crops. Such cases will therefore be underrepresented on the Project scale. Planting cover crops as soon as is feasible (i.e., directly after cash crop harvest/termination) helps counteract this imbalance.

Attribute Dependencies

- Harvest Date
- Termination Date

3.2.5 Harvest Date

Logic

- For cash crops:
  - Check NASS Usual Harvest Dates (see `IndigoCarbon_US-1_2023_0069` for details) using the associated cash crop type.
  - For crop/state combinations that are missing, use proxy states (Table 1).
  - Convert the Usual Harvest date to day-of-year in the given year being gap-filled.
  - If NASS lookup fails, then try the remote sensing check for harvest date.
  - If both NASS and remote sensing estimates fail, then fail to fill the harvest date estimate.
- For cover crops:
  - For cover crops grown in non-irrigated fields:
    - Check NRCS Zone in which the field is located (Zones are defined in the NCRS Cover Crop Termination Guidelines, see `IndigoCarbon_US-1_2023_0069` for details) and use the offset from planting date associated with that Zone:
      - Zone 1: 35 days before planting of next cash crop
      - Zone 2: 15 days before planting of next cash crop
      - Zone 3: 1 day before planting
      - Zone 4: 1 day before planting
  - For cover crops grown in irrigated fields:
    - 1 day before planting of the next cash crop

Rationale

- It is assumed that cover crops are terminated more often than they are harvested for grain or stover. This is supported by the NRCS Cover Crop Termination Guidelines (`IndigoCarbon_US-1_2023_0069`) which recommends when to terminate a cover crop based on geography and irrigation status. The logic for cover crop termination date, which is based on the NRCS document, is also used here for cover crop harvest date.
- The use of NASS Usual Harvest Dates for cash crops conforms with Step 2 of the data hierarchy in the SEP v1.1, 6.1 for quantitative data (Section 6.1, pp 70).
- The use of NASS Usual Harvest Dates is consistent with the QA/QC ranges developed for planting and harvest events (see `IndigoCarbon_US-1_2023_0032`).

Attribute Dependencies

- Cash Crop Type
- Cash Crop Planting Date

3.2.6 Harvest Method

Logic

- Replace the attribute with "HAY" if the cash crop type is alfalfa, else replace the attribute with "GRAIN ONLY".

Rationale

- Default replacement of Harvest Method only occurs on cash crops.
- It is agronomically reasonable to assume a grass cash crop such as alfalfa has been planted with the intent to harvest it for hay or haylage; here it is assumed to be hay.
- For all other cash crops it is agronomically reasonable to assume the harvest is for grain only. (Other options include baled straw, grain with residue removal, haylage, and silage.)

Attribute Dependencies

- Cash Crop Type

3.2.7 Harvest Yield

Logic

- Replace with NASS estimate, which depends on whether the field is irrigated.
- If NASS estimate is not available, fall back to SME estimate.
- Calculate SME estimate by taking the average of the minimum range value and the max range value. See `IndigoCarbon_US-1_2023_0032` for SME harvest range values.

Rationale

- The use of NASS values conforms with Step 2 of the data hierarchy for quantitative data in the SEP, v1.1 (Section 6.1, pp 70).
- The approach is consistent with the QC ranges developed for yield values described in `IndigoCarbon_US-1_2023_0032`.

Attribute Dependencies

- Crop
- Harvest method
- Irrigation start date

3.2.8 Termination Type

Logic

- This attribute is always replaced with "TERMINATION".

Rationale

- The options for Termination Type are "TERMINATION", "LEFT IN FIELD", or "CROP FAILURE". Termination events are most commonly associated with cover crops, and it is assumed that cover crops are terminated more often than they are left in the field or experience failure.

Attribute Dependencies

- None

3.2.9 Termination Method

Logic

- This attribute is always replaced with "HERBICIDE".

Rationale

- The options available for termination method in the Grower Survey are Herbicide, Mowing, Roller Crimper, Tillage, and Winter Kill. If tillage is selected, growers are required to add a tillage event to match it. From a modeling perspective, all the non-tillage methods will be fairly equivalent because they do not remove residue.
- For cover crops, which are presumably the most common case of termination, the USDA ERS report *Cover Crop Trends, Programs, and Practices in the United States* (Wallander et al. 2021) indicates that chemical termination is most common. Chemical termination was used on about $\frac{2}{3}$ of cover crop fields whose cash crop was soybean, corn-for-grain, and cotton, and on a plurality of fields whose cash crop was corn-for-silage. Other cash crops are likely similar but were not commonly enough cover-cropped to have a reliable estimate in the survey. Therefore, it is reasonable to assume a cover crop terminated without harvest was terminated by chemicals.
- We expect cash crops to be terminated only rarely and will never assume a termination of a cash crop by default. In the unlikely event we know the grower terminated a cash crop but do not know how, chemical termination is an agronomically reasonable assumption.

Attribute Dependencies

- None

3.2.10 Termination Date

Logic

- If Termination Method = Winter Kill
  - Replace the attribute with December 31 if there is exactly one December 31st between the terminated planting and next cash crop planting.
  - Otherwise, fail.
- Else, if there is a next cash crop planting, set the date relative to the next cash crop planting
  - For cash crops: set the date to 1 day before planting of next cash crop
  - For cover crops grown in non-irrigated fields:
    - Check NRCS Zone in which the field is located (Zones are defined in the NCRS Cover Crop Termination Guidelines, see `IndigoCarbon_US-1_2023_0069`) and use the offset from planting date associated with that Zone:
      - Zone 1: 35 days before planting of next cash crop
      - Zone 2: 15 days before planting of next cash crop

        - Zone 3: 1 day before planting
        - Zone 4: 1 day before planting
    - For cover crops grown in irrigated fields:
        - 1 day before planting of next cash crop
- Else, if there is no next cash crop planting, fail.

Rationale

- This approach is based largely on NRCS cover crop termination guidelines, which are designed to avoid cases where the cover crop is unsuccessful (i.e., impedes the cash crop) and thus crop insurance is negated. The central concept behind the NRCS termination date guidelines is water availability to the cash crop, with drier climates needing to terminate sooner to allow more water to be available at planting. For irrigated systems it is assumed that sufficient water will be available to support the cash crop and thus termination can wait until the day before planting.
- This approach does not allow crop growing seasons to overlap and is thus logically consistent with QA/QC (see Gate 2, Level 2 description in `IndigoCarbon_US-1_2023_0032`).

Attribute Dependencies

- Termination Method
- Cash Crop Planting Date

3.2.11 Residue Burned

Logic

- This attribute is always replaced with "NOT BURNED".

Rationale

- Nationally, most crops' residues are burned only very rarely. For example, Table 5-32 of the US EPA *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2019* (U.S. Environmental Protection Agency, 2021) shows 6% of rice production area, and no more than 3% of any other listed crop, burned in 2014.
- Regional variation was checked using a county-level version of the same 2014 crop burning dataset used for the national inventory. Results indicated that even crops that are "commonly burned" in certain regions are still mostly *not* burned (i.e., rates of burning in these regions varied between rare and very rare, not between prevalent and not prevalent). See the appendix at the end of this document for details of the analysis.

Attribute Dependencies

- None

3.2.12 Chemical Application Date

Logic

- Using the crop type, look up the date of application in table 2
- The replacement date will be offset from either "days before harvest/termination" or "days after planting".
- Attempt to replace the attribute using that date:
    - Ensure that the application occurs within the planted period. For example, if "90 days before harvest" precedes planting, use the planting date.
    - Make sure there are no other chemical application events taking place on that day. If there are, march backward in time (i.e., earlier), to find the next earliest day on which there is no chemical application event.

– Ensure the application is not within 28 days of harvest, to be consistent with the SME check in Gate 2 Level 4. If the date does fall within 28 days of harvest, shift backward in time to the first available date at least 29 days before harvest.

Rationale

- The Chemical Application Regime table was developed based on:
  - *USDA Crop Conservation Trends Report* (Baranski et al. 2018) (see Figure 15)
  - Indigo SME guidance
- Based on model sensitivity, the date of N application has a small effect on modeled results of SOC (see `IndigoCarbon_US-1_2023_0072`).
- Model testing using DayCent-CR showed that outcomes of SOC-based emission reductions were nominally affected by switching from a single (100%) fall fertilizer application per year to a single pre-plant application per year. The model test was conducted across 377 US locations, spanning a variety of climates and soil types. The shift in application timing for these sites resulted in an average change in SOC of 0.07 tCO2eq ac-1 y-1.
- The default equations used to calculate N_2O emissions operate at the annual scale only and are thus insensitive to multiple applications per year.
- The logic is consistent with any range checks in QA/QC (Gate 2).

Attribute Dependencies

- Cash Crop Type
- Cash Crop Planting Date
- Harvest Date
- Termination Date

3.2.13 Chemical Application Product

Logic

- Using the crop type, look up the chemical application product in table 2.

Table 2: Chemical Application Regime

Crop	Application Date	Product
Corn	5 weeks post-plant (35d)	UAN
Cotton	1 month post-plant (30d)	UAN
Winter wheat, Spring Wheat, Barley, Rye, Triticale, Canola	3 months prior to harvest or termination (90d)	UAN
Sorghum, Sunflowers, Millet, Flax, Oats	Planting	UAN
Lentils, chickpeas, dry beans, field peas	Planting	MAP
Soybeans	Planting	10/34/0

Rationale

- Replacement fertilizer product for each crop was informed by:
  - A USDA ERS report (*Fertilizer Use and Price*, 2019) citing that nitrogen solutions (e.g., UAN) are the most common form of nitrogen fertilizer applied in the US.
  - Indigo SME guidance
- According to Indigo SMEs, N-fixing legumes do not typically need N fertilizer but do often receive P fertilizer that contains small amounts of N, namely MAP, DAP, or "10-34-00" forms. Both MAP and DAP are among the most common according to the ERS Report.

Attribute Dependencies

- Cash Crop Type

3.2.14 Chemical Application Rate

Logic

- Look up the NASS QuickStats (see `IndigoCarbon_US-1_2023_0069` for details) average annual N fertilizer rate by cash crop and state (i.e., *NASS_365* in the database*)*.
  - For crop/state combinations that are missing, use proxy states (Table 2 presented in Missing Chemical Application Events section).
- Scale the NASS annual rate according to the length of the agronomic year.
  - *NASS_N = NASS_365 * (n_days_in_this_agyear / 365)*
- Obtain the total amount of N already applied via existing fertilizer events (i.e., *Already_Applied_N)*.
- Calculate the remaining amount of N available to gap-fill with as:
  - *Gapfill_N = max(0.0, NASS_N – Already_Applied_N)*
- Replace the attribute using *Gapfill_N*. If $Gapfill_N = 0$, the event can be ignored.

Rationale

- The impact of fertilizer N rate on modeled SOC is high (see `IndigoCarbon_US_1_2023_0072`), thus a standard survey of application rates is used from NASS Quick Stats. Use of NASS Quick Stats data is consistent with Step 2 of the SEP v1.1, 6.1 data hierarchy for quantitative data (Section 6.1, pp 70). To fully conform with NASS Quick Stats rates, the replacement approach does not over-apply beyond the NASS rate. If the grower has already exceeded this amount, no additional N fertilizer is applied.

Attribute Dependencies

- Cash Crop Type

3.2.15 Organic Amendment Type

Logic

- Check the most commonly applied manure type for the region according to manure management data from the USDA-ERS ARMS crop production practice survey (see `IndigoCarbon_US-1_2023_0069`).
- Replace the attribute with the most common manure type for this field's ARMS Farm Production Region.

Rationale

- The use of ARMS manure data is consistent with requirements for qualitative data in the SEP v1.1 6.1 data hierarchy (Section 6.1, pp 70).
- Because organic amendment type is primarily determined by the type of organic amendment that is locally available, it is agronomically reasonable to assume a grower will use the same amendment type that is most common in their region.

Attribute Dependencies

- None

3.2.16 Organic Amendment Rate

Logic

- Look up the average ARMS (see `IndigoCarbon_US-1_2023_0069`) annual manure application rate for this year's cash crop (i.e., *ARMS_OM*) using the Tables 3 & 4. Units are in tons of total applied amendment per acre per crop year (i.e. includes solid plus liquid).

  - If a regional value is available for the given crop x state, use it.
  - Else use the regional average across crops (crop = "ALL_CROPS" in Table 4) for the region that contains the field.
- Obtain the total amount of OM already applied via existing organic amendment events (i.e., *Already_Applied_OM).*
- Calculate the remaining amount of OM available to gap-fill with as:
  - *Gapfill_OM = max(0.0, ARMS_OM – Already_Applied_OM)*
- Replace the attribute using *Gapfill_OM*. If *Gapfill_OM* = 0, the event can be ignored.
- *For Triticale, use the same values as winter wheat.*

Table 3: ARMS Average Manure Application Rates by Crop and Region

Crop	Region	Tons dry matter applied per acre per crop year
barley	Appalachian	9.09
barley	Lake States	19.77
barley	Mountain	15.67
barley	Northeast	15.04
barley	Northern Plains	9.85
barley	Pacific	6.87
barley	TOTAL	15.14
corn	Appalachian	3.04
corn	Corn Belt	4.66
corn	Lake States	10.96
corn	Mountain	15.67
corn	Northeast	10.1
corn	Northern Plains	4.51
corn	Southeast	2.37
corn	Southern Plains	11.24
corn	TOTAL	9.95
cotton	Appalachian	1.95
cotton	Mountain	11.01
cotton	Pacific	5.62
cotton	Southeast	1.84
cotton	Southern Plains	6.22
cotton	TOTAL	2.62
durumwheat	Northern Plains	1.76
durumwheat	TOTAL	1.76
oats	Corn Belt	7.01
oats	Lake States	13.76
oats	Northeast	11.03
oats	Northern Plains	2.07
oats	TOTAL	10.79
peanuts	Southeast	1.59
peanuts	Southern Plains	1.66
peanuts	TOTAL	1.59
rice	Delta	1.93
rice	Southern Plains	1.47
rice	TOTAL	1.9
sorghum	Mountain	1.09
sorghum	Northern Plains	7.28
sorghum	Southern Plains	2.75
sorghum	TOTAL	5.54
soy	Appalachian	2.35
soy	Corn Belt	4.3
soy	Delta	1.42
soy	Lake States	6.95
soy	Northeast	5.1
soy	Northern Plains	3.43
soy	TOTAL	4.92
springwheat	Lake States	9
springwheat	Mountain	8.64
springwheat	Northern Plains	9.9
springwheat	Pacific	3.82

Table 3: ARMS Average Manure Application Rates by Crop and Region

Crop	**Region**	**Tons dry matter applied per acre per crop year**
springwheat	TOTAL	7.48
winterwheat	Appalachian	8.29
winterwheat	Corn Belt	3.91
winterwheat	Mountain	5.72
winterwheat	Northern Plains	4.03
winterwheat	Pacific	8.21
winterwheat	Southern Plains	6.53
winterwheat	TOTAL	5.55
ALL_CROPS	Appalachian	2.18
ALL_CROPS	Corn Belt	4.55
ALL_CROPS	Delta	1.59
ALL_CROPS	Lake States	11.89
ALL_CROPS	Mountain	3.56
ALL_CROPS	Northeast	9.86
ALL_CROPS	Northern Plains	2.98
ALL_CROPS	Pacific	5.64
ALL_CROPS	Southeast	1.78
ALL_CROPS	Southern Plains	6.19
ALL_CROPS	TOTAL	2.53

Table 4: State to region mapping.

State	**Region**
KY	Appalachian
NC	Appalachian
TN	Appalachian
VA	Appalachian
WV	Appalachian
IA	Corn Belt
IL	Corn Belt
IN	Corn Belt
MO	Corn Belt
OH	Corn Belt
AR	Delta
LA	Delta
MS	Delta
MI	Lake States
MN	Lake States
WI	Lake States
AZ	Mountain
CO	Mountain
ID	Mountain
MT	Mountain
NM	Mountain
NV	Mountain
UT	Mountain
WY	Mountain
CT	Northeast
DE	Northeast
MA	Northeast

Table 4: State to region mapping.

State	Region
MD	Northeast
ME	Northeast
NH	Northeast
NJ	Northeast
NY	Northeast
PA	Northeast
RI	Northeast
VT	Northeast
KS	Northern Plains
ND	Northern Plains
NE	Northern Plains
SD	Northern Plains
CA	Pacific
OR	Pacific
WA	Pacific
AL	Southeast
FL	Southeast
GA	Southeast
SC	Southeast
OK	Southern Plains
TX	Southern Plains

Rationale

- The use of ARMS data is consistent with Step 2 of the SEP v1.1, 6.1 data hierarchy for quantitative data (Section 6.1, pp 70). To fully conform with ARMS rates, the replacement approach does not over-apply beyond the ARMS rate. If the grower has already exceeded this amount, no additional organic amendment is applied.

Attribute Dependencies

- Cash Crop Type

3.2.17 Organic Amendment Date

Logic

- If the date is being replaced because a grower-specified date was out of range:
  - Obtain the next harvest date; QA/QC will enforce that this be < 28 days after the originally specified organic amendment date (see `IndigoCarbon_US_1_2023_0032`).
  - If no existing organic amendment applications occur on the date 29 days prior to harvest, replace the attribute with the date 29 days prior to harvest.
  - Else if there is an existing application on the date 29 days prior to harvest, walk backwards one day at a time until a date is found not having an organic amendment application.
- If the date is newly created to replace a completely missing organic amendment event:
  - Replace the attribute with the date 28 days after harvest. The missing event logic will enforce that no other organic amendment events exist in the crop growing season, so there is no need to check for conflicting events on this date.

Rationale

- The timing of organic amendment application is highly variable and often opportunistic. Growers apply amendments when labor is available and when the field is accessible for spreading. However, given that the model is insensitive to the date of organic matter addition (`IndigoCarbon_US_1_2023_0072`), it is more

important to capture the correct total C and N rate applied. The default replacement value thus assumes a single application based on the ARMS annual rate (see *Organic Amendment Rate*).

- When a whole event is missing, assuming it occurred during the late fall after the previous season's harvest is agronomically reasonable.
- When the attribute is out of range, the grower reported applying organic amendments less than 28 days before harvest. Replacing the date with one earlier in the growing season is agronomically reasonable and ensures that the nutrients from the application are counted as available to the same crop the grower reported applying it to, whereas replacing the date with one after harvest would cause the nutrients to be allocated to a subsequent crop.

Attribute Dependencies

- Harvest Date

3.2.18 Tillage Date

Logic

- For new tillage events only:
  - Obtain cash crop planting date.
  - Find the most recent crop harvest or termination date.
  - If the most recent harvest or termination date is $\geq$ 3 days prior to planting, set the default tillage date to 3 days prior to planting.
  - Else, set the tillage date to the day of the previous harvest or termination.

Rationale

- This approach is supported by previous modeling studies that treat tillage as closely associated with planting and harvest operations when precise dates are unknown:
  - Lutz et al. (2019): Global multi-crop simulations of tillage scenarios using the LPJmL model; with-tillage simulations have tillage "twice a year, at sowing and after harvest";
  - Andales et al. (2000): Simulations of NE Iowa soybean using CROPGRO; "Tillage operations were specified: field cultivation before planting (disk chisel and moldboard systems only), row planting on the actual planting dates, and two post-emergence cultivation operations for weed control (all three tillage systems). In the simulations, Field cultivation was assumed to occur the day before planting while the postemergence cultivation operations were done when soil conditions were favorable (e.g., soil was well drained, no rainfall 2 days prior to cultivation)."
  - Stehfest et al. (2007): Global crop production simulations using DayCent; assumes tillage is "one single ploughing just before planting."

Attribute Dependencies

- Cash Crop Planting Date
- Harvest Date
- Termination Date

3.2.19 Tillage Method

Logic

- If tillage date falls between planting and harvest or termination date of any crop (cash or cover), assume the tillage is for in-season weed control and replace the attribute with "ROW CULTIVATION".
- Else set the tillage method to "VERTICAL TILL".

Rationale

- The choice of a low impact tillage method ("VERTICAL TILL") is based on a model experiment of DayCent-CR that showed low impact tillage was the most conservative when gap-filling tillage events in Baseline scenarios and medium impact tillage was the most conservative when gap-filling tillage events in with-project scenarios. The experiment compared average emission reductions across 36 model runs using different tillage methods. The model runs spanned 12 soil types and three geographic locations.
- This is logically consistent with the assumption of the SEP that lower impact tillage sequesters more SOC than higher impact tillage, as well as model sensitivity (`IndigoCarbon_US-1_2023_0072`).
- Despite the goal of using the same value in both baseline and with-project scenarios, it is necessary here to choose a default value that will be conservative at the Project scale. This is done based on the assumption that more baseline values than with-project values will be replaced because growers are less likely to have records on-hand from years further in the past. Thus, a conservative default value of tillage method for the entire Project is the one that is conservative in the baseline scenario. According to the model experiment this is a low-intensity tillage method. From the tillage methods identified as low-intensity, "Vertical Till" was chosen as the lowest intensity of the conventional seedbed preparation methods, while "Row Cultivation" is better matched to the degree of disturbance produced by tillage for weed control during the growing season.

Attribute Dependencies

- Tillage Date
- Cash Crop Planting Date
- Cover Crop Planting Date
- Harvest Date
- Termination Date

3.2.20 Irrigation Start Date

Logic

- Repace this attribute with the cash crop planting date.

Rationale

- The start date of irrigation refers to when the auto-irrigation routine in the model begins monitoring soil moisture for the need to irrigate (i.e., when soil moisture drops below a certain threshold). Once soil moisture falls below the threshold, water is applied. It is agronomically reasonable, and supported by Indigo SMEs, to assume that a grower will begin to monitor the need to irrigate once a crop is planted.

Attribute Dependencies

- Cash Crop Planting Date

3.2.21 Irrigation End Date

Logic

- Replace this attribute with the date 30 days prior to the harvest or termination date.
- If the planted period is shorter than 30 days, then set the date to the harvest or termination date.

Rationale

- By default, irrigation is only associated with cash crop growing seasons, never cover crop growing seasons.
- This default replacement approach is supported by Indigo SMEs who recommend irrigation be stopped roughly one month prior to harvest to allow a cash crop to dry down.

Attribute Dependencies

- Harvest Date
- Termination Date

3.2.22 Grazing Hours Per Day

Logic

- Use 24 hours when attribute is null.

Rationale

- The only type of grazing in the Project at this time is cropland grazing, and we expect the majority of cropland grazing will be 24 hours per day.
- 24 hours is a conservative value that will lead the the highest animal-grazing-days, and associated emissions, when used in the project scenario. In cases where 24 hours is gapfilled in the baseline only, the leakage provision will prevent the gapfilled value from being anti-conservative.

Attribute Dependencies

- None

3.3 References

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3.4 Appendix: Residue Burned Analysis Summary

Residue burning prevalence by crop and county

When a harvest event is missing its "Residue Burned" attribute, we always assume no burning. This is accurate across the whole Project because residue burning is very rare on a nationwide basis and occurs on $\leq$% of production area in every crop evaluated (EPA National Greenhouse Gas Inventory, . To evaluate whether we needed to account for regional variation in burning prevalence (i.e. assume burning for some crops in some locations), we examined county-level burning prevalence using a dataset developed for the US EPA's 2014 National Greenhouse Gas Inventory.

The full analysis was performed in an interactive Python session using a Jupyter code notebook. A PDF version of the output takes up 94 pages, so this document summarizes the analysis and its major findings.

Method

Area burned per crop

To find the area burned of each crop, we used a national remote-sensing dataset (referred to below as the "Pouliot data") containing burning incidence by crop in 2014, which was developed for the US National Emissions Inventory and was downloaded from the address https://catalog.data.gov/dataset/development-of-the-crop-residue-and-rangeland-burning-in-the-2014-national-emissions-inven as "Pouliot_A-4b8m_Dataset_20170307.xlsx" The data.gov page describes this dataset as follows:

"This workbook contains all the activity data, emission factor data, and ancillary data used to compute crop residue burning and rangeland emissions for the 2014 NEI as described in the journal article (see citation) in addition to the data associated with Figures 1-5.

> "This dataset is associated with the following publication: Pouliot, G., V. Rao, J. McCarty, and A. Soja. Development of the crop residue and rangeland burning in the 2014 National Emissions Inventory using information from multiple sources. JOURNAL OF THE AIR & WASTE MANAGEMENT ASSOCIATION. Air & Waste Management Association, Pittsburgh, PA, USA, 67(5): 613-622, (2017)."

Similar datasets for years other than 2014 do not appear to be immediately available, but crop burning is relatively consistent from year to year (McCarty et al. 2009, "The spatial and temporal distribution of crop residue burning in the contiguous United States"; https://doi.org/10.1016/j.scitotenv.2009.07.009) and therefore we assume these estimates are applicable to other recent years.

See Pouliot et al 2017 for details on the development and limitations of this dataset, but in summary it is based on remote sensing of burn events using HMS (a composite data product created from multiple satellite sources), cross-referenced with the USDA Cropland Data Layer (CDL) to filter out non-agricultural areas and to identify the crop whose residue is assumed burned. Also notably, the size of each burn event is assumed from typical field size in the state (one of 40, 60, 80, or 120 acres) rather than computed from the satellite data; this avoids errors from potentially inaccurate pixel counts but means counties with only a few acres of a given crop can be estimated as >100% burned.

Total crop area

To find the total area of each crop grown in each county, we used two datasets:

1. A subset (1499 counties) of the same 2014 CDL data used by Pouliot et al, processed internally by Indigo into field-level crop assignments, which we then summed to give total acreage grown in each county. Note that the selection of which counties to include and the field-level assignment was done with a focus on row crops while accepting lower coverage of grasslands, forage crops, and hay/pasture. These provide high spatial resolution and since they come from the same dataset the county-level crop totals from CDL are the best denominator to convert burned area to percent burned, but only include a subset of the 2216 counties included in the Pouliot data.
2. The 2014 NASS Quickstats data gives county-by-county acres harvested for major commodity crops. These are available for all counties where a given crop has major production but includes fewer crops than CDL.

In counties where both datasets were available, NASS acreage was usually somewhat higher than CDL acreage, but the correlation between them was high and qualitative conclusions were similar when using either one.

Summary of findings

Fifteen crops have low burning rates everywhere they are grown (never more than 25% of acres burned in any county with >2000 acres of the crop grown, and conversely all counties with more than 25% burned can all be attributed to single-digit numbers of burn events in counties where the crop is only grown on a very few acres), and can safely be assumed not-burned by default (table 5).

Four crops do not have enough information in this dataset to make an accurate determination (table 6). All of these are crops that are not currently included in Indigo's program and are likely under counted in the data we used for this analysis, making it hard to determine whether apparent high burn rates in some counties are true common management or an artifact of under counting the total crop area. If we extend the program to include these in the future, the burning assumption should be rechecked using a dataset better optimized to identify grassland crops.

Table 5: Crops not associated with burn events.

Alfalfa	Barley	Beans/Peas
Canola	Corn	Cotton
Flax	Oats	Rice
Rye	Sorghum	Soybeans
Sugar Beets	Sunflowers	Wheat (all varieties)

Table 6: Crops with limited burn event data

Fallow	Grass/Pasture
Other hay/non-alfalfa	Sugarcane

4 Summary of Changes

Section	**Date**	**Summary of Changes**
Default replacement	June 20, 2024	Added details on default replacement for grazing hours per day